

MONITOR

Introduction to Monitor

- ☑ A display screen to provide “output” to the user. It is where you view the information your are working on.



Introduction to Monitor

- ☑ Monitors often use internal charges of tens of thousands of volts that can be persist long after the power disconnected.
- ☑ High risk of electric shock.

Types of Displays/Monitors

- ☑ These days, monitors available in markets are of many different kinds;
- ☑ unlike older days, when people do not have options other than CRT Monitors. PC monitors come in two different flavor.

CRT Monitors

- ☑ Cathode Ray Tube, or in simpler words CRT, are oldest among all available models these days.
- ☑ The traditional, glass-screen, television-set-like monitor.

CRT Monitors

- ☑ The best thing in CRTs is that they are easily available and are in **cheaper in price**.
- ☑ These monitors have high dynamic range and remarkable **color and resolutions** with **excellent viewing angles**.
- ☑ One of the drawbacks of these monitors is their **weight** and another is their **size**.
- ☑ They **consume more power**, like many other old electrical and electronic appliances.
- ☑ That is why, experts on energy do not advice to buy CRT.

CRT Monitor



LCD Monitors

- ☑ Stands for **liquid crystal display**. The newer, flatter type of computer screen.
- ☑ Unlike CRT monitors, these monitors are **compact and slim**.
- ☑ These monitors do consume low and almost have no dependency on backlight technology.
- ☑ Due to its quality of **consuming low power** and **compact in shape and size**, it has been well adapted in the time when **energy efficiency is the main concern**.
- ☑ In spite of all these technological modest characteristics, these monitors have **limited viewing angles, colors, and contrasts**.

LCD Monitor

Issues like bleeding and distorting brightness from edges and some more related issues have been reported in some models.



Plasma Monitors

- ❑ Plasma screen and/or plasma monitors are considered as **high contrast screen** with bright, vibrant colors and brightness that claims to make your visual experience worthwhile.
- ❑ It works on plasma discharge on almost ideally flat panel of glass.
- ❑ Plasma monitors are in use , mainly because of their **excellent and remarkable viewing angles, color saturation, and contrasts**.
- ❑ In comparison with LCD, Plasma monitors have less blocky-looking picture. However, Plasmas are **heavy in weight and available in larger dimensions** .

Plasma Monitor



SED Monitors

- ☑ SED is abbreviated form of Surface-conducted electron-emitted display.
- ☑ These are high resolution and flat panel display screens. Some of these display units are even more than 40 inches in diagonal measurements.
- ☑ SED consumes less energy in comparison with CRT and it gives higher resolution picture.



- ❑ Other than, all mentioned types of monitors , there are ;
 - ❑ Organic light-emitting diode(OLED) monitors,
 - ❑ Digital Light Processing Monitors,
 - ❑ touch screen monitors,
 - ❑ digital displays and
 - ❑ projector displays used for commercial and special purposes.

Monitors can also be classified in terms of color capabilities, which separates monitors into three classes:

- ❑ **monochrome** : Monochrome monitors actually display two colors, one for the background and one for the foreground. The colors can be black and white, green and black, or amber and black.
- ❑ **gray-scale**: A gray-scale monitor is a special type of monochrome monitor capable of displaying different shades of gray.
- ❑ **color**: Color monitors can display anywhere from 16 to over 1 million different colors. Color monitors are sometimes called RGB monitors because they accept three separate signals -- red, green, and blue.

Monitor...

- ☑ The other most important aspect of a monitor is its **screen size**. Like televisions, screen sizes are measured in diagonal inches, the distance from one corner to the opposite corner diagonally.
- ☑ A typical size for small VGA monitors is 14 inches. Monitors that are 16 or more inches diagonally are often called *full-page monitors*.

Monitor...

- ☑ In addition to their size, monitors can be either portrait (height greater than width) or landscape (width greater than height). Larger landscape monitors can display two full pages, side by side.
- ☑ The screen size is sometimes misleading because there is always an area around the edge of the screen that can't be used.
- ☑ monitor manufacturers must now also state the viewable area -- that is, the area of screen that is actually used.

Health and Safety Concerns

- ☑ Display screen equipment has been associated with a range of health problems.

Eye and vision problems

- ☑ Work on display screen equipment can be visually very demanding. The more intensive the work, the more likely people are to suffer vision disorders.

Symptoms

- ☑ eyestrain.
- ☑ burning or sore eyes.
- ☑ headaches,
- ☑ irritated eyes.
- ☑ blurred vision.
- ☑ change in colour perception; problems with colour recognition.
- ☑ tiredness and irritability.
- ☑ discomfort when wearing contact lenses.

Likely causes

- ❑ incorrect positioning of the equipment and documents.
- ❑ inappropriate and/or inadequate lighting.
- ❑ poorly designed workstations.
- ❑ long periods of work at the screen.
- ❑ uncorrected vision defects.
- ❑ low humidity.
- ❑ poorly designed or poor quality screens.
- ❑ lack of adequate maintenance of equipment.

Musculo-skeletal problems

- ☑ Musculo-skeletal problems associated with display screen equipment work are generally caused by poor posture, poor layout of the workstation and repetitive movements.
- ☑ They include problems in the back, neck, shoulders, arms and hands etc.

Recommended Advices

- ☑ Display Screen Equipment (DSE) used in the Service must meet the following minimum standard(s).

1.The **display screen** must have:-

- ☑ characters that are **well-defined**, clearly formed, of adequate **size** and with adequate **spacing**;
- ☑ a **stable image** with no flickering;
- ☑ brightness and contrast **controls** which are easily adjustable by the user;
- ☑ be **free of reflective glare/reflections** which are liable to cause discomfort to the user.

Cont'd

2. The work desk or work surface must have:

- ☑ a sufficiently large, low reluctant, surface which will allow a **flexible arrangement** of the screen, keyboard, documents and related equipment.

3. The lighting must be:-

- ☑ suitable for Display Screen Equipment (DSE) work, i.e. **not too bright, but bright enough** to allow the user to read comfortably

Monitor Connections

- ☑ Monitors have different types of connections to hook it to a computer.

VGA

- ☑ The Video Graphics Array or Adapter connection has been around since 1987, and it can be recognized by the 15-pin configuration.
- ☑ Most VGA ports on the computer are blue.

VGA cable of a video monitor



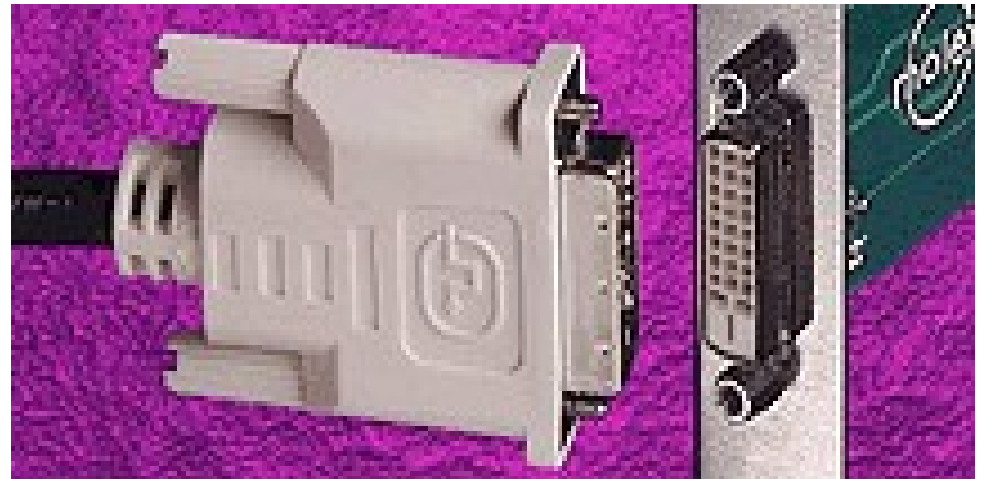
S-Video(Separated Video)

- ❑ One different type of connection for a monitor that gives better picture quality than VGA is S-Video.
- ❑ This is a **5-pin** setup that typically has **yellow or black** ports.
- ❑ S-Video connector and cable



DV-I(Digital Video Interface)

- ☑ Since the mid-2000's, the Digital Video Interface connection has become a common port on computers and monitors.
- ☑ There are **24 pins** on this connection, and the port is usually **white** or off-white.
- ☑ DVI Connector and Cable



HDMI (High-Definition Multimedia Interface)

- ☑ For a true digital connection with precise color, definition and sound, use an HDMI cable.
- ☑ This is a port that resembles a USB port, but it is slightly tapered or narrow.



Video Ports on a video card



Troubleshooting the Video System/Monitors

- ☑ It is extremely dangerous to remove the back cover of a monitor,
 - ☑ particularly a CRT (Cathode Ray Tube) type, as voltages in excess of 25,000 volts can be present, which could remain for some considerable time after the power has been removed.
- ☑ Although the front face of a CRT has a thick glass wall, the rear neck of the tube is thin and easily broken

Troubleshooting the Video System/Monitors

- ❑ If no image is seen, the signals being sent by the computer may be **incompatible**.
- ❑ CRT Monitors will only accept a certain range of vertical and horizontal scan ranges. Unsuitable scan rates will produce unstable or incoherent images.
- ❑ Flat Panel Monitors only display sharp images at a specific resolution, called the **native resolution**.
- ❑ This varies between monitors, but is usually consistent for a given size of screen. For a 15 inch screen, this typically would be **1028 * 768 pixels**.
- ❑ If a monitor is presented with any resolution other than the native resolution, the image will appear less sharp or distorted or both.

Troubleshooting the Video System/Monitors

- ☑ Preventive maintenance for a monitor is pretty simple - just keep the case **clean and free of obstructions**.
- ☑ For CRT monitors, clean the screen with a soft cloth just dampened with **water** and mild **detergent** or isopropyl **alcohol**.
- ☑ DO NOT use anything so wet that liquid may seep inside of the monitor around the edge of the CRT.
- ☑ You could end up with a very expensive repair bill when the liquid decides to short out the main circuit board

Troubleshooting Tips for Monitors

1) The picture does not appear

- ☑ Check to make sure the **signal cable** is firmly connected in the socket.
- ☑ Check to see if the **computer system's power** is ON.
- ☑ Check that the **Brightness Control** is at the appropriate position, not at the minimum.

Troubleshooting Tips for Monitors

2) The Screen is not synchronized

- ☑ Check to make sure the **signal cable is firmly connected** in the socket.
- ☑ Make sure the **signal timing** of the computer system is within the specification of the monitor.

Troubleshooting Tips for Monitors

3) The position of the screen is not in the center

- ☑ Adjust the H-Size, H-Phase or V-Size, V-Center controls.
- ☑ Check if the signal timing of the computer system is within the specification of the monitor.

Troubleshooting Tips for Monitors

4) The screen is too bright or too dark

- ☑ Check if the **Brightness or contrast** control is at the appropriate position, not at the maximum or minimum.
- ☑ Check if the **specified voltage** is applied
- ☑ Check if the **signal timing** of the computer system is within the specification of the monitor.

Troubleshooting Tips for Monitors

5) The screen is shaking

- ☑ Move all objects that emit a **magnetic field**, such as a motor or transformer, away from the monitor.
- ☑ Check if the **specified voltage** is applied.
- ☑ Check if the **signal timing** of the computer system is within the specification of the monitor.

BIOS(Basic Input Output System)

- ☑ BIOS is a program built into personal computers that starts the operating system when you turn on your computer.
- ☑ **The BIOS built-in software contains specific information about the computer and motherboard it was designed for.**
- ☑ It is also referred to as system **firmware**. BIOS is part of your computer's hardware and is separate from Windows.
- ☑ BIOS doesn't need to be managed and you don't need to change any settings.

BIOS..

- ☑ Advanced users might choose to change certain settings, such as the order the computer searches devices when starting.
- ☑ During the first boot of the computer, after the motherboard and required hardware and devices are physically installed, BIOS begins to record all that it finds, and sends it to be stored in CMOS RAM.
- ☑ If on subsequent boots, devices or other important details in the computer change, so does the record within CMOS.
- ☑ Therefore, ROM BIOS and CMOS RAM must both be present on the motherboard, and be able to work together for the system to function.

The initial responsibility of the BIOS is to:

- ☑ Provide a basic set of instructions to boot the computer.
- ☑ Run **POST (Power On Self Test)**
- ☑ Effect a workable routine with either an FDD boot-up disk or the HDD, through an interface, so that the computer can be accessed.
- ☑ Provide an interface between the motherboard and all other hardware, via a library of interrupt handlers, for the operating system, so they can all work together.

Error messages and solutions

- ❑ **POST Error Message**-A POST error message is an error message displayed on the monitor during the Power On Self Test if the BIOS encounters some kind of problem while starting the PC.
- ❑ A POST error message will only display on screen if the computer is capable of booting this far. If the POST detects an error before this point, a beep code or POST code will be generated instead.
- ❑ **CMOS Battery State Low or CMOS battery failed**
The battery voltage is low and the battery should be replaced. Make sure you have the correct battery

BIOS...

- ☑ **Disk Boot failure-insert system disk and press enter**
- ☑ If your computer has just been formatted, your computer might require you to insert the boot CD to be able to run it.
- ☑ You might get the error message **Disk Boot Failure-Insert disk and press enter** displayed on the screen every time you will try to run it.

BIOS...

☑ This might be linked to:

1. The boot options not correctly set in BIOS
2. A non bootable CD is found in your CD drive
3. You do not possess any bootable device
4. You have installed a new hard disk
5. The hard disk is not properly plugged in
6. Your system files have been infected

Solutions

- ☑ **Ensure that your drives are empty**
 - ☑ If there is a non-bootable CD or disk in the drives, your computer might be attempting to boot the program with this disk.
 - ☑ Hence, ensure that the floppy disk drive, the CD/R and DVD drives are all empty before restarting your computer and trying to run it again.

Solutions

☑ BIOS set up

- ☑ If this has not solved the problem, it might be linked to an incorrectly set up BIOS.
- ☑ 1.To enter BIOS, press the F8 key when the Windows screen shows up upon starting the computer
- ☑ 2.Then all you have top do is to prioritize the boot sequence to Disk C:

New hard drive configuration

- ☑ If you have just installed a new hard drive, you should insert the system installation CD and restart your computer again so that it reboots with the CD.

☑ **Hard drive not properly plugged in**

- ☑ If your hard drive is not properly plugged in, this can also lead to a rebooting error message. Switch off your computer and remove it from the power supply before checking the cables attached to it. Ensure that the motherboard is correctly installed. Once you have done this, try running your system again.

☑ **Restore the corrupted system files**

Beeps Error Details

- ❑ **1 long, 2 short**-Video adapter Error.
 - ❑ Either the Video adapter is bad or is not seated properly.
 - ❑ Also, check to ensure the monitor cable is connected properly.
- ❑ **Repeating, endless loop**
 - ❑ Memory Error
 - ❑ Check for improperly seated or missing RAM
- ❑ **High-frequency beeps while running**
 - ❑ Overheated CPU
 - ❑ Check the CPU fan for proper operation
 - ❑ check case for proper airflow.
- ❑ **Repeating High/Low**
 - ❑ CPU error
 - ❑ Either the CPU is not seated properly or the CPU is damaged.
 - ❑ May also be due to excess heat.
 - ❑ Check the CPU fan (or BIOS settings for proper fan speed).

Advanced BIOS Features

☒ **Virus Warning (Enabled)**

- ☒ When enabled, the BIOS performs a Pre-Operating System Check for viruses that could possibly modify the boot sector of the HDD.
- ☒ If any modification or scheduled modification is encountered, the boot process will come to a halt and the DOS prompt will generate the message
- ☒ Boot sector is about to be modified, do you wish to continue? Y/ N.

- ☑ This BIOS-based virus check is very helpful because it catches boot sector viruses and it should be enabled under normal operating conditions.
- ☑ **CPU Internal Cache (Enabled)**
 - ☑ All modern CPUs have at least one level of internal cache memory (SRAM).
 - ☑ The function of the internal (level 1, L1)
 - ☑ cache is to store data and instructions that have been read from the main memory and are written back to the cache for faster access in case they are requested again.

☑ **Boot-Up Floppy Disk (Disabled)**

- ☑ Some older floppy drives have only 40 tracks (360 / 720 kBytes) whereas the new drives have 80 tracks (1.44 / 2.88 MBytes).
- ☑ **Enabled** means that the system will check which type of drive is installed in the system.
- ☑ This is meaningless if the system is using new drives.
- ☑ It takes up boot time and may decrease the life span of the floppy drive because no diskette is inside the drive while checking.
- ☑ This setting should be disabled.

Video BIOS Shadow (Disabled)

- ☑ When the BIOS is shadowed it means it has been copied to RAM so that DOS programs and some older [Windows drivers](#) run a little faster.
- ☑ In legacy graphics cards , copying the video BIOS into the system memory could increase the graphics performance of the system.
- ☑ For all modern graphics cards, the setting should be disabled because there is no performance increase and the possibility to encounter errors because of occupying other necessary system resources.

☑ System BIOS Shadow (Disabled)

- ☑ Similar to video BIOS shadowing, after the BIOS is loaded, there is no reason to shadow it in the main memory for modern OS.

How do I access my computer's BIOS?

- ☑ Procedures vary depending on the BIOS manufacturer.
- ☑ Usually, you must press a key (such as F2, F12, Delete, or Esc) or a key combination immediately after you turn on your computer before Windows starts.
- ☑ For more information, check the information that came with your computer or go to the computer manufacturer's website.

Warning!

- ☑ Be careful when changing BIOS settings.
- ☑ The BIOS interface is designed for advanced users, and it's possible to change a setting that could prevent your computer from starting correctly.
- ☑ Updating BIOS should only be done if necessary (to solve a compatibility problem, for example).
- ☑ It can be a complicated process, and if an error occurs, your computer could be rendered inoperable.
- ☑ Be sure to follow the manufacturer's instructions exactly.

Thank You

End of the Chapter